



Project of DSCE-499 for 2nd 2025-2026

A Data Visualization Experiment on the Superstore Retail Sales Dataset

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Research Questions:

This project sets out to answer the following questions through data visualization:

1. How do sales change over time (trend and seasonality)?
2. Which product category generates the highest sales?
3. Which region performs best in terms of sales?
4. What is the relationship between sales and profit, and do discounts affect profitability?

Each visualization in the Output section is designed to answer one or more of these questions.

1. Dataset

This project uses a real-world, publicly available retail dataset known as the "Sample Superstore" dataset. It is one of the most widely used datasets for data-visualization and business-intelligence training. It originates as official sample data distributed by Tableau Software and has since been redistributed openly on Kaggle and GitHub.

- Dataset name: Sample Superstore (US retail orders).
- Original provider: Tableau Software sample data.
- Public copy used in this project: github.com/leonism/sample-superstore
- Direct CSV: raw.githubusercontent.com/leonism/sample-superstore/master/data/superstore.csv

The dataset records individual customer orders placed at a US retail store between January 2015 and December 2018. After removing blank trailing rows during cleaning, it contains 9,994 valid order records — the standard published size of this dataset. Each record includes:

- Order Date / Ship Date — order and shipping dates (time dimension).
- Region, State, City — geographic location of the order.
- Segment — customer type (Consumer, Corporate, Home Office).
- Category, Sub-Category, Product Name — the product hierarchy.
- Sales — order revenue in US dollars.
- Quantity — number of units ordered.
- Discount — discount rate applied (0 to 0.8).
- Profit — profit on the order in US dollars (can be negative).

This dataset is robust enough to support the analytical goals of the experiment: it combines a continuous time dimension, several categorical dimensions (region, category, segment), and multiple numeric measures (sales, quantity, discount, profit). This makes it possible to study trends over time, compare groups, examine relationships between variables, and analyze distributions. A descriptive summary of the numeric fields is shown in Table 2

2. Tools Selection

This project follows the Programming Stream. All data cleaning, exploration, and visualization were carried out in Python using the following libraries:

- **pandas — for loading CSV files, cleaning the data, parsing dates, and grouping/aggregation.**
- **NumPy — numeric operations that support pandas.**
- **Matplotlib — the core plotting engine, used to create the time-series trend chart.**
- **Seaborn — a statistical visualization library built on Matplotlib, used for the bar charts, scatter plot, correlation heatmap, and ranking chart.**

Python with Matplotlib and Seaborn was chosen because these libraries integrate directly with pandas data frames, offer fine-grained control over chart formatting, and produce publication-quality figures with concise code. We chose the Programming Stream rather than Power BI because our team is more proficient with Python, while we are not yet experienced with Power BI.

3. Implementation

The implementation is contained in a single Python script (analysis. ipynb) that runs in three stages: (1) loading the dataset, (2) cleaning and preparing it, and (3) generating six visualizations saved as PNG files. Key portions of the source code are shown below.

3.1 Loading and cleaning the data

```
import pandas as pd, numpy as np

import matplotlib.pyplot as plt, seaborn as sns

sns.set_theme(style='whitegrid')

# Load the real Superstore dataset

df = pd.read_csv('superstore.csv')

# Remove blank trailing rows -> 9,994 valid records

df = df.dropna(subset=['Sales', 'Order Date', 'Region', 'Category'])

# Parse dates and derive time fields

df['Order Date'] = pd.to_datetime(df['Order Date'], format='%m/%d/%Y')

df['month'] = df['Order Date'].dt.to_period('M').astype(str)

df['year'] = df['Order Date'].dt.year
```

3.2 Example visualization code

```
# Monthly sales trend (Matplotlib line chart)

monthly = df.groupby('month')['Sales'].sum().reset_index()

plt.plot(monthly['month'], monthly['Sales'])

# Sales by category (Seaborn bar chart)

sns.barplot(data=df, x='Category', y='Sales', palette='viridis')

# Profit vs sales (Seaborn scatter plot)

sns.scatterplot(data=df, x='Sales', y='Profit', hue='Category')

# Correlation heatmap of numeric features

corr = df[['Sales', 'Quantity', 'Discount', 'Profit']].corr()

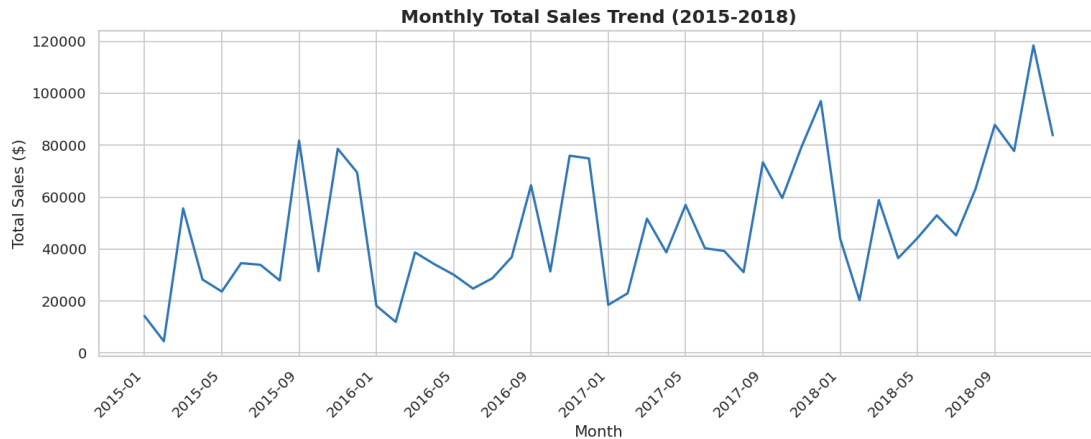
sns.heatmap(corr, annot=True, cmap='coolwarm', center=0)
```

The complete script (analysis. ipynb) and the cleaned dataset (superstore_clean.csv) are provided with this report so the experiment can be reproduced end-to-end.

4.Output

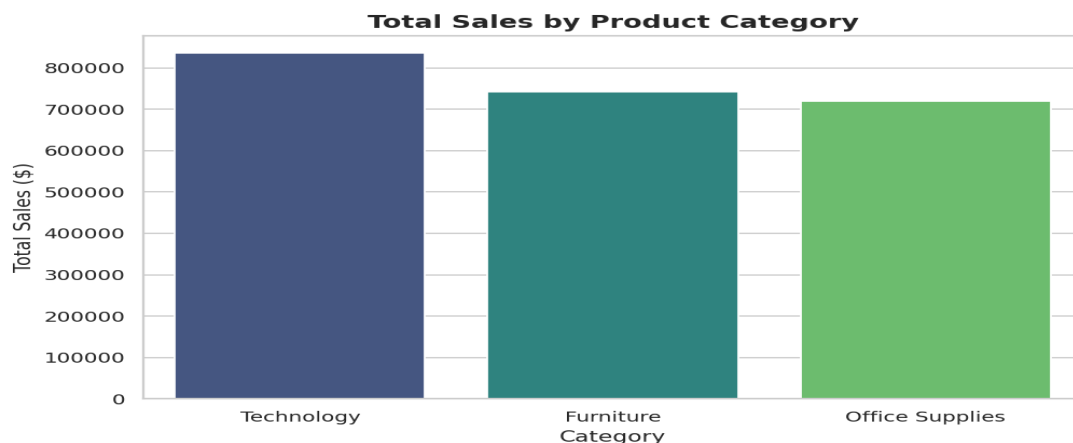
Running the script produces the six figures below directly from the real data. Each visualization answers a specific analytical question.

Figure 1 — Monthly Sales Trend



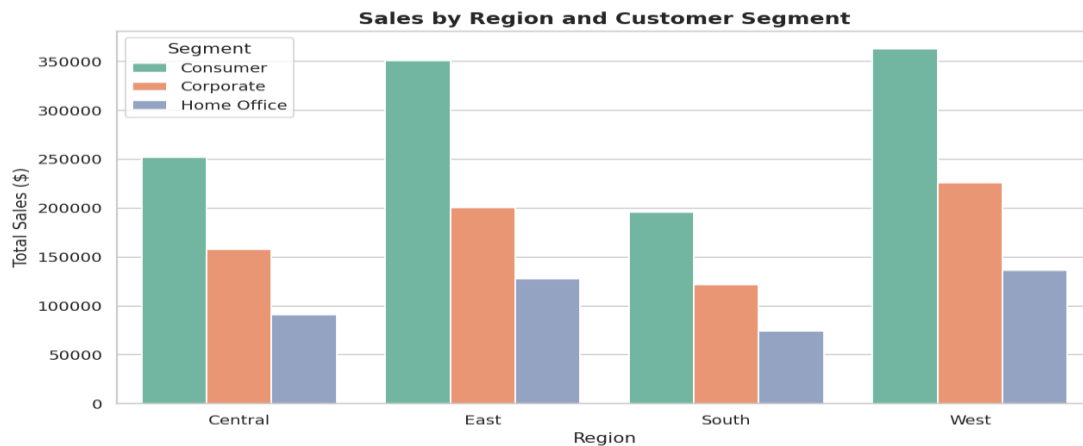
Sales show a clear upward trend over the four years, together with a strong seasonal pattern: sales peak sharply toward the end of each year (November–December) and dip early in the year. This is typical retail seasonality and demonstrates how a Matplotlib line chart reveals both trend and seasonality.

Figure 2 — Sales by Category



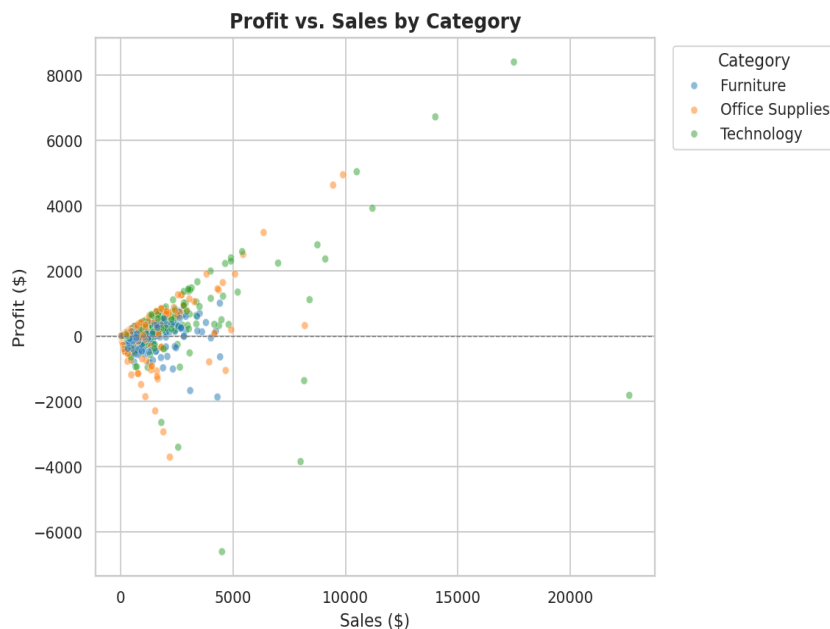
The three categories contribute fairly close totals, with Technology generating the highest revenue, followed closely by Furniture and Office Supplies. The bar chart makes category-level comparison immediate.

Figure 3 — Sales by Region and Segment



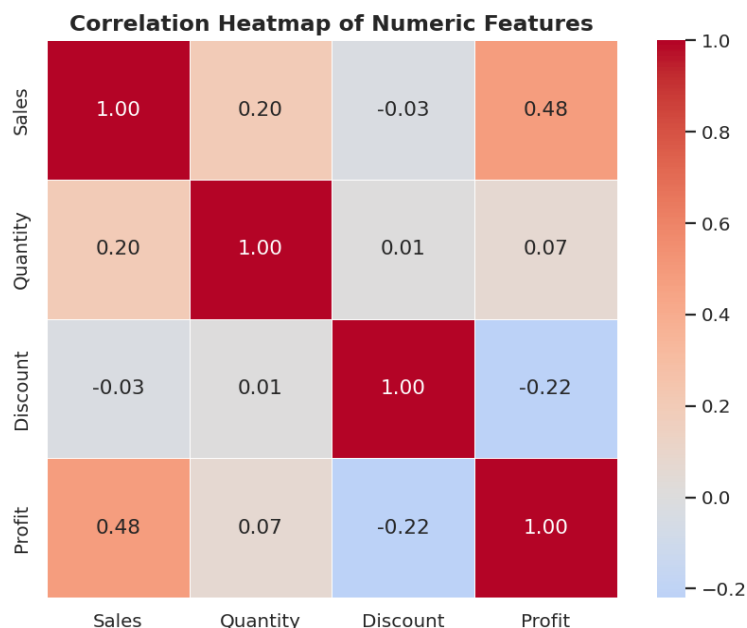
The West and East regions generate the most sales, while within every region the Consumer segment dominates, followed by Corporate and Home Office. The grouped bars compare two categorical dimensions at once.

Figure 4 — Profit vs. Sales



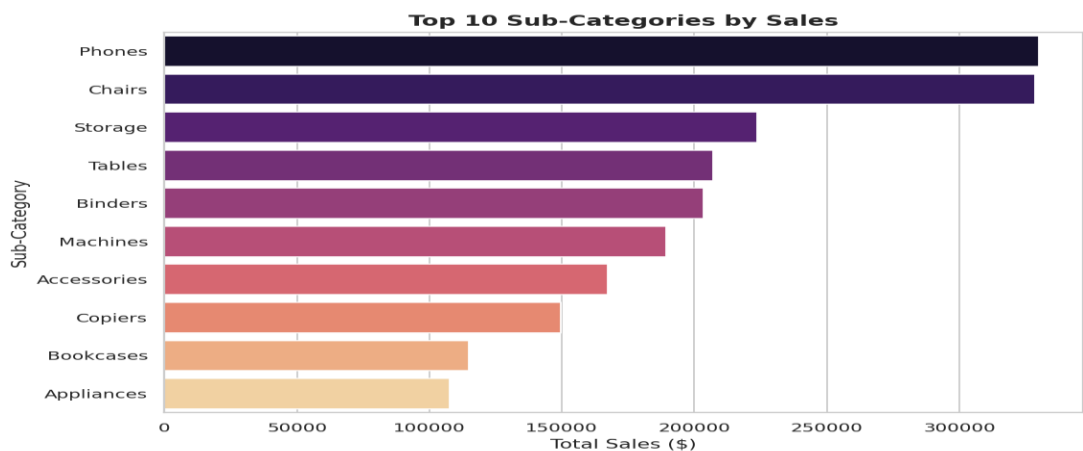
Although higher sales generally bring higher profit, a large number of orders fall below the zero line and are sold at a loss — about 18.7% of all orders. Loss-making orders are common across all categories, especially where heavy discounts are applied. This is one of the most important business insights from the dataset.

Figure 5 — Correlation Heatmap



Sales and profit are moderately positively correlated (about 0.48), while discount and profit are negatively correlated (about -0.22). This quantifies a key finding: larger discounts tend to erode profit, even when they help drive sales volume.

Figure 6 — Top 10 Sub-Categories by Sales



Phones and Chairs are the top-selling sub-categories, followed by Storage and Tables. This horizontal ranking chart highlights which specific product lines drive the most revenue.

Conclusion:

This experiment applied a complete data-visualization workflow to a real-world retail dataset (Sample Superstore) using the Python programming stream. After cleaning the data down to 9,994 valid orders, six complementary visualizations were produced with Matplotlib and Seaborn, covering trend, comparison, relationship, and distribution analysis. The analysis directly answers the four research questions:

1. Sales over time: sales grew across 2015–2018 and show strong year-end seasonality, peaking every November–December.
2. Top category: Technology is the highest-selling category, followed closely by Furniture and Office Supplies.
3. Best region: the West region achieved the strongest sales performance, and the Consumer segment dominates in every region.
4. Sales, profit & discounts: sales and profit are positively correlated, but discounts negatively impact profitability — about one in five orders (18.7%) is sold at a loss. At the product level, Phones and Chairs generate significant revenue.

Across the period the store generated about \$2.30 million in sales and \$286 thousand in profit. In summary: Technology is the highest-selling category; the West region achieved the strongest performance; discounts negatively impact profitability; and Phones and Chairs generate significant revenue. Together these visualizations turn raw transaction records into clear, decision-ready insight, fulfilling the analytical goals of the project.